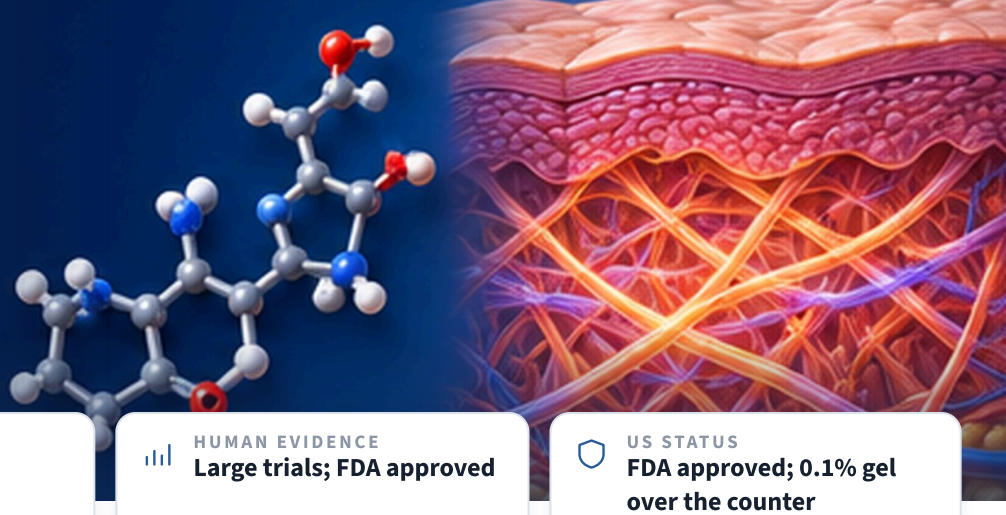
SKIN 09

Adapalene

Retinoid (vitamin A-like drug) put on the skin that binds retinoic acid receptors in the cell nucleus



CLASS Topical retinoid

STUDIED FOR Acne

HUMAN EVIDENCE Large trials; FDA approved

US STATUS FDA approved; 0.1% gel over the counter

STUDIED FOR

- **Acne** — HUMAN
- **Moderate-severe acne** — HUMAN

HOW IT WORKS IN THE BODY

Adapalene ▶ Retinoic acid rec. ▶ Skin cell turnover ▶ Acne lesions

- **Binds retinoic acid nuclear receptors,** not the cytosolic receptor protein
- **Modulates** cell differentiation, keratinization and inflammation
- **Meaning for acne** of these findings is unknown, per the label

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Differin approved 1996; OTC since 2016
- 0.3%/0.1% gel trial, 653 people, 12 weeks
- Clear or almost clear: 21% / 16% vs 9%

NOT YET ESTABLISHED

- Use in pregnancy
- Circulating metabolites

QUICK PRACTICAL INFO

HALF-LIFE
Mean 17.2 hours (range 7 to 51) after skin use

ROUTES STUDIED
On the skin (gel, cream, solution)

PHARMACOKINETICS
Low blood levels; mostly gone by 72 hours; biliary excretion

SOLD AS
OTC 0.1% gel (Differin); prescription 0.3% gel, cream, combinations

MONITORED IN TRIALS
Skin irritation, sun sensitivity

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved 1996; 0.1% gel switched to OTC (2016)
FDA	Differin Epiduo OTC approved (May 2026)
EMA	Authorised nationally in all 27 EU countries
WADA	Not on the 2026 list

REFERENCES

1. FDA, Differin gel 0.3% prescribing information (DailyMed), Dec 2023
2. FDA, Prescription to Nonprescription Switch List, Jun 2026
3. Cunliffe et al., Br J Dermatol, adapalene vs tretinoin pooled analysis, Oct 1998

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